HDL-P, Total

Test Details

METHODOLOGY

Nuclear magnetic resonance (NMR)

RESULT TURNAROUND TIME

2 - 3 days

Turnaround time is defined as the usual number of days from the date of pickup of a specimen for testing to when the result is released to the ordering provider. In some cases, additional time should be allowed for additional confirmatory or additional reflex tests. Testing schedules may vary.

Use

HDL-P, a measurement of total HDL particle number concentration, may be a better marker of residual risk than chemically measured high-density lipoprotein cholesterol (HDL-C, the so-called “good” cholesterol) or apolipoprotein A-1 (apoA-1, the major protein on HDL), ie, there may be a more consistent inverse association between cardiovascular endpoints and HDL-P compared with HDL-C.

Direct quantification of HDL-P concentration by NMR may be useful to refine cardiovascular risk and to evaluate novel HDL-directed therapies. Further studies are needed to clarify the role of HDL-P in clinical practice.

Limitations

This test was developed and its performance characteristics determined by Labcorp. It has not been cleared or approved by the Food and Drug Administration.

Specimen Requirements

Specimen

Spun NMR LipoTube (preferred), serum from a plain red-top tube, plasma from a lavender-top (EDTA-no gel) **or** green-top (heparin-no gel) tube. Keep refrigerated and **ship with frozen cool packs.**

Volume

2 mL

Minimum Volume

1 mL

Container

NMR LipoTube (black-and-yellow-top tube) is the preferred container; plain red-top tube, lavender-top (EDTA-no gel) tube **or** green-top (heparin-no gel) tube

Collection Instructions

Collect specimen in NMR LipoTube (black-and-yellow-top tube), which is the preferred container. Plain red-top, green-top (heparin-no gel) or lavender-top (EDTA-no gel) tubes also are acceptable. Serum or plasma drawn in gel-barrier collection tubes other than the NMR LipoTube should **not** be used. **The LipoTube is the only acceptable gel-barrier tube.** Gently invert tube eight to 10 times to mix contents and allow specimen to clot for 30 minutes upright at room temperature prior to centrifugation (plasma tubes should not clot). Centrifuge specimen within two hours of collection at 1600 to 1800 xg for 10 to 15 minutes to separate serum/plasma from the red cells and to avoid red cell contamination during shipment. If the sample cannot be centrifuged immediately, the sample should be refrigerated (at 2°C to 8°C) and centrifuged within 24 hours of collection. **Note:** Centrifuging the specimen while still cold may negatively affect the migration of the gel to the serum/red cell interface and may increase the likelihood of specimens being contaminated with red cells during shipment. All specimens should be centrifuged by the client **prior** to shipment to Labcorp to ensure sample integrity. Do **not** open NMR LipoTube (black-and-yellow-top). Immediately after centrifugation, pipette separated red-top serum or green-top/lavender-top plasma into a transport tube and label accordingly (serum, heparin plasma, EDTA plasma). Keep samples refrigerated until shipment to the laboratory, and **ship with frozen cool packs.**

Storage Instructions

Refrigerate all acceptable tube types as soon as possible after centrifugation and within 24 hours of collection. Keep refrigerated prior to shipment, and **ship on frozen cool packs**. Do **not** store at room temperature. Do **not** freeze the sample. Sample is stable refrigerated for 14 days.

Patient Preparation

Patient should be fasting for 12 to 14 hours.

Causes for Rejection

Unspun specimens; plasma/serum contaminated with red cells; citrated plasma (light blue-top tube); gross hemolysis; specimen received in inappropriate container; specimen stored at room temperature for more than a total preanalytical time of 30 hours; specimen more than 14 days old